Luke Chipman

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Education

B.A.Sc. Mechanical Engineering (Materials Option), 4.0 GPA, Queen's University Sep 2024 - May 2028

Professional Experience

Mechanical Engineering Intern - Twente Additive Manufacturing May 2026 - Present

- Independently led design, fabrication, and testing of a 6-tool automatic tool changer enabling two concrete 3D printing robots to operate collaboratively
- Worked on site installing and assembling 2 concrete 3D printers
- Led transition of manufacturing to an overseas machine shop, producing technical drawings and verifying part quality and tolerances
- Created documentation for all mechanical parts and systems designed

Mechanical Engineering Intern - Twente Additive Manufacturing 2025 May 2025 - Aug 2025

- Used Rhino/Grasshopper and robot studio to program and operate 7 axis concrete 3D printers
- Developed and operated live demos for potential clients by creating parametric CAD for 3D prints
- Prototyped experimental woodworking attachments for the printers

Founder - Luke Chipman Designs Feb 2020 - Aug 2024

- Scale 3D printed architectural models of houses, accurately showing the surrounding topography, exterior, and floor plan
- Designed aluminum frames to hold glass diffusion filters used for cinema cameras
- Designed and sold 3D printed props on Etsy, generating over $10k in sales
- Sold over 700 files of prop designs

Extracurricular and Project Experience

Mechanical Director - Queen's Aerospace Design Team May 2026 - Present

- Overseeing the mechanical sub teams in design and manufacturing a fixed wing, composite UAV for the SAE Aero 2026 competition
- Advising on mechanical design and ensuring integration across the team

Aero - Structures Manager - Queen's Aerospace Design Team May 2025 - Apr 2026

- Led a 10-person team to design and fabricate a fully carbon fibre UAV to compete in the AIAA DBF 2026 competition
- Used composite lay-ups, waterjet cut carbon internal structure, FDM 3D printing for fabrication
- Conducted FEA on design elements
- Integrated with aerodynamics, electrical, and software teams

Lunar Greenhouse Harvesting Mechanism - Canadian Space Agency Jan 2025 - May 2025

- Designed an end effector for a robotic arm for manipulation of small-scale plants
- Attended weekly meetings with mentors from the Canadian Space Agency for design and professional development

Achievements

Canadian Engineering Competition, Junior Design - 3rd place March 2026

- Competed against regional finalists from across Canada to design, build, and present a prototype to solve an engineering problem. Qualified by winning the Queen's and Ontario Engineering Competition

MechMania Judges' Choice Award April 2026

- Competed against 275 Mechanical Engineering students in a 4-day design challenge

Technical Skills

Hardware: 3D printing, Composite Layups, Laser Cutting, Machine shop experience, Soldering, Welding
Design: SolidWorks, Onshape, Rhino/Grasshopper, Ansys, Rhino Robot, Robot Studio, Technical Drawings
Programming: Python, C, C++, MATLAB
Productivity: Excel, Word, Microsoft Office, PowerPoint